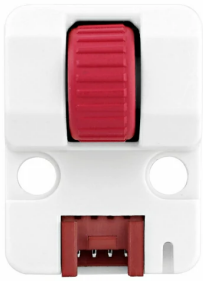
Unit Scroll

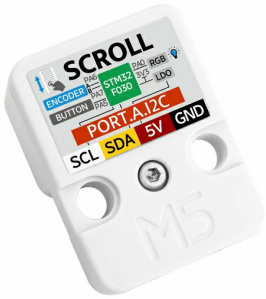
SKU:U186



















Description

Unit Scroll is a rotary encoder expansion unit with a scroll-wheel form factor. It integrates a 12-bit through-shaft pulse encoder, a button input, and a WS2812C programmable RGB LED. The built-in STM32F030 microcontroller runs firmware for encoder pulse signal acquisition, allowing users to directly obtain encoder values via I2C read operations, greatly simplifying the usage process. This unit is suitable for control scenarios such as value scaling adjustment and volume control.

Features

- Based on STM32 microcontroller
- AB phase output (12 pulses / rotation)
- Built-in button
- Colorful WS2812C RGB LED indicator
- I2C communication interface

Includes

- 1 x Unit Scroll
- 1 x HY2.0-4P Grove Cable (20cm)

Applications

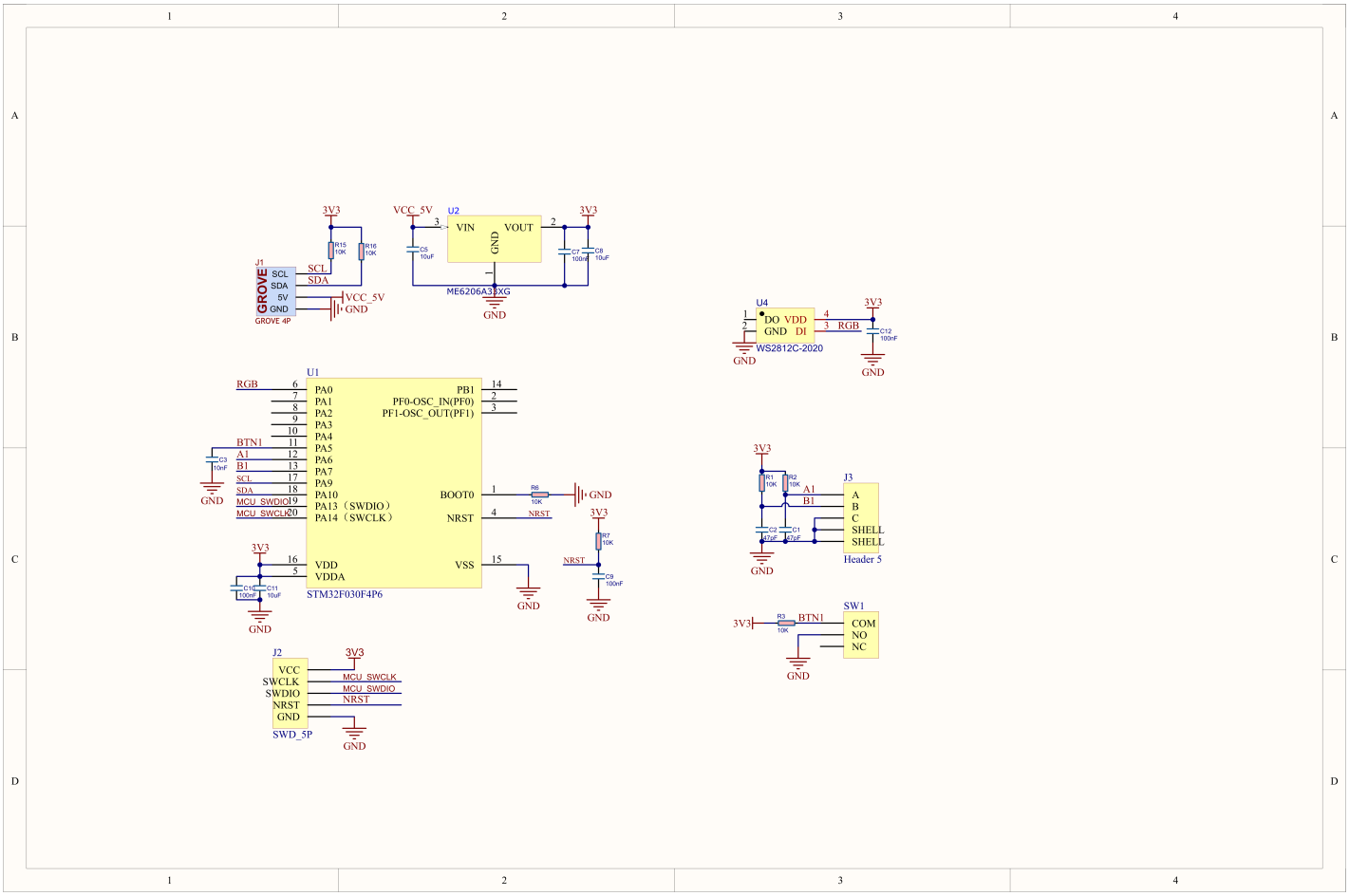
- Industrial automation
- Smart home
- Robotics
- Human–machine interaction

Specifications

Specification	Parameter
MCU	STM32F030F4P6@32-bit ARM Cortex-M0
Rotary Encoder	EC10E
Encoder Output	AB quadrature output
Encoder Resolution	12 pulses / rotation
RGB LED	1x WS2812C
Button	1x Button
I2C Address	0x40
Operating Temperature	0 ~ 40°C
Product Size	32.0 x 24.0 x 18.9mm
Product Weight	7.5g
Package Size	138.0 x 93.0 x 19.9mm
Gross Weight	12.7g

Schematics

- Unit Scroll Schematics PDF



PinMap

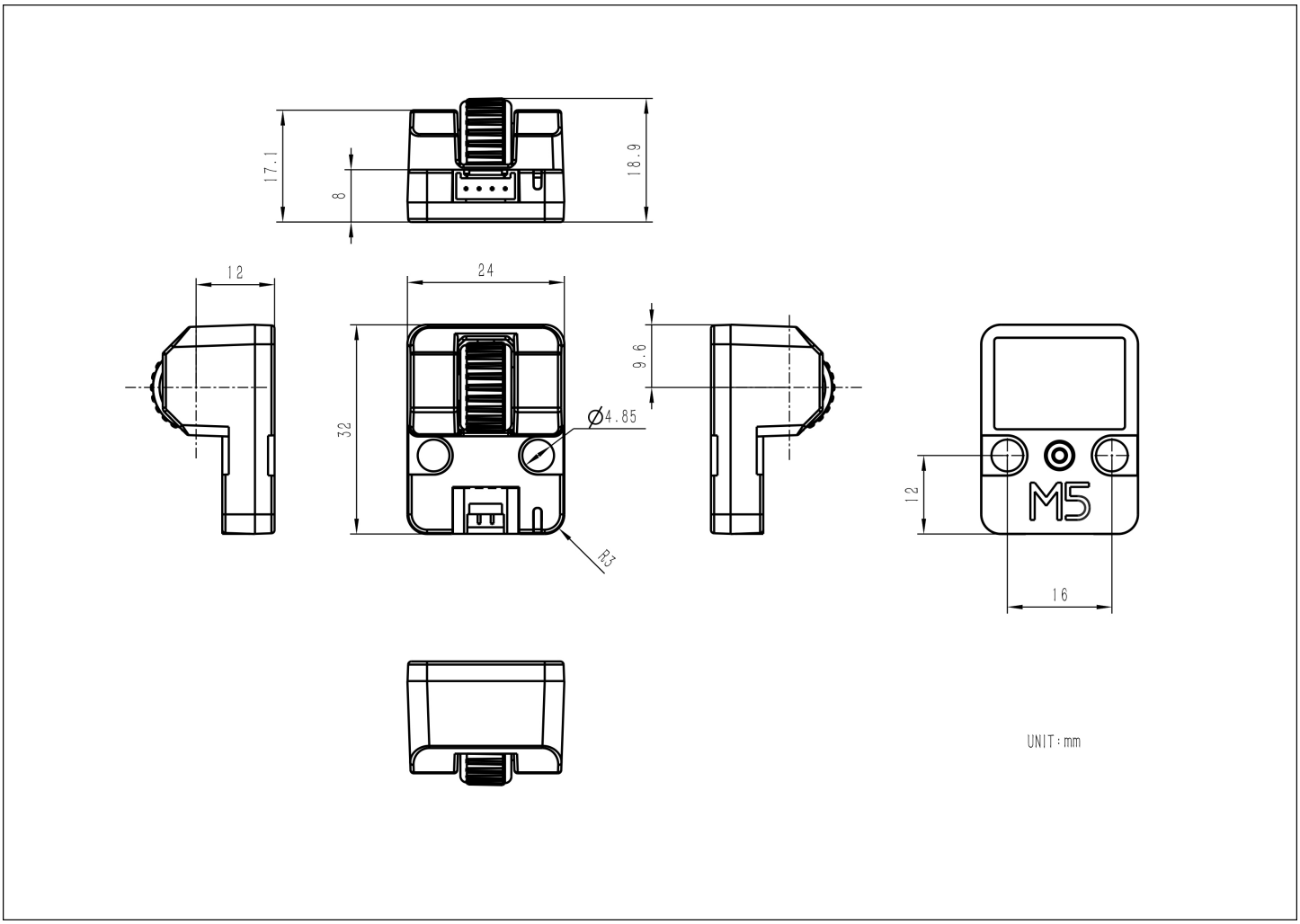
Unit Scroll

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

STM32F030F4P6 Internal Peripheral Pin Connections

STM32F030F4P6	PA6	PA7	PA9	PA10	PA0	PA5
Encoder	A1	B1				
I2C			SCL	SDA		
WS2812C RGB LED					RGB	
Button						BTN1

Model Size



Structure

- Unit Scroll Structure Files

Datasheets

- Rotary Encoder EC10E

Softwares

Arduino

- Unit Scroll Arduino Library

Internal Firmware

- Unit Scroll Internal Firmware

Protocol

M5Stack Unit Scroll I2C Protocol																	V1 (FW Version) 2024/2/29
REG MAP (0x40)		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	note
Encoder	0x10 W/R	Encoder -byte0	Encoder -byte1															Encoder = Encoder-byte0 + Encoder-byte1 * 256
Button	0x20 R	Button																0~1
RGB	0x30 W/R	NULL	R	G	B													R/G/B
Reset	0x40 W	Reset																write 1, Reset Encoder value(10H) and Inc Encoder Value(50H)
Inc Encoder	0x50 R	Inc Encoder -byte0	Inc Encoder -byte1															Inc Encoder = Inc Encoder-byte0 + Inc Encoder-byte1 * 256 will set to 0 after read
Encoder AB or BA	0xF0 W/R												Encoder AB or BA					Encoder AB or BA: 0: AB(default) 1: BA
Bootloader Version	0xF0 R													Bootloader Version				Value: bootloader version
Firmware Version	0xF0 R															FW Version		Value: firmware version
I2C Addr	0xF0 R																I2C Addr	Value: I2C Address

UiFlow2

- Unit Scroll UiFlow2 Docs

Video

- Unit Scroll Feature Introduction

U186 Scroll Unit 视频.mp4

- UiFlow2 Unit Scroll

Product Comparison

Product Compare			
	Unit Scroll	**Unit Encoder**	**Unit 8Encoder**
Form	Wheel	Knob	Knob
Channels	1	1	8
Resolution	12 pulses/rotation	30 pulses/rotation	15 pulses/rotation
Position	AB Quadrature Output	AB Quadrature Output	AB Quadrature Output
Communication Mode	I2C (0x40)	I2C (0x40)	I2C (0x41)